



Supplementary Information for

Reconciling modern machine learning practice and the classical bias-variance trade-off

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S1 Approximation theorem

Suppose the training data $(x_1, y_1), \dots, (x_n, y_n)$ are sampled independently by drawing x_i uniformly from a compact domain in $\mathbb{R}^d$, and assigning the label $y_i = h^*(x_i)$ using a target function $h^* \in \mathcal{H}_\infty$. Let $h \in \mathcal{H}_\infty$ be another hypothesis that interpolates the training data $(x_1, y_1), \dots, (x_n, y_n)$. The following theorem bounds the error of h in approximating h^* .

Theorem 1. *Fix any $h^* \in \mathcal{H}_\infty$. Let $(x_1, y_1), \dots, (x_n, y_n)$ be independent and identically distributed random variables, where x_i is drawn uniformly at random from a compact cube¹ $\Omega \subset \mathbb{R}^d$, and $y_i = h^*(x_i)$ for all i . There exists absolute constants $A, B > 0$ such that, for any interpolating $h \in \mathcal{H}_\infty$ (i.e., $h(x_i) = y_i$ for all i), so that with high probability*

$$\sup_{x \in \Omega} |h(x) - h^*(x)| < A e^{-B(n/\log n)^{1/d}} (\|h^*\|_{\mathcal{H}_\infty} + \|h\|_{\mathcal{H}_\infty}).$$

Proof sketch. Recall that the fill κ_n of the set of points $x_1, \dots, x_n$ in Ω is a measure of how well these points cover Ω : $\kappa_n = \max_{x \in \Omega} \min_{x_j \in \{x_1, \dots, x_n\}} \|x - x_j\|$. It is easy to verify (e.g., by taking an appropriate grid partition of the cube Ω and applying the union bound) that with high probability $\kappa_n = O(n/\log n)^{-1/d}$.

Consider now a function $f(x) := h(x) - h^*(x)$. We observe that $f(x_i) = 0$ and by the triangle inequality $\|f\|_{\mathcal{H}_\infty} \leq \|h^*\|_{\mathcal{H}_\infty} + \|h\|_{\mathcal{H}_\infty}$. Applying Theorem 11.22 in [15] to f yields the result. $\square$

The minimum norm interpolating function $h_{n,\infty}$ has norm no larger than that of h^* (by definition) and hence achieves the smallest bound in Theorem 1. While these bounds apply only in the noiseless setting, they provide a justification for the inductive bias based on choosing a solution with a small norm. Indeed, there is significant empirical evidence that minimum norm interpolating solutions generalize well on a variety of datasets, even in the presence of large amounts of label noise [1].

S2 Experimental setup

To demonstrate the double descent risk curve, we train a number of representative models including neural networks, kernel machines and ensemble methods on several widely used datasets that involve images, speech, and text.

Table S1: Descriptions of datasets. In experiments, we use subsets to reduce the computational cost.

Dataset	Size of full training set	Feature dimension (d)	Number of classes
CIFAR-10	$5 \cdot 10^4$	1024	10
MNIST	$6 \cdot 10^4$	784	10
SVHN	$7.3 \cdot 10^4$	1024	10
TIMIT	$1.1 \cdot 10^6$	440	48
20-Newsgroups	$1.6 \cdot 10^4$	100	20

¹Same argument can be used for more general domains and probability distributions.

Datasets. Table S1 describes the datasets we use in our experiments. These datasets are for classification problems with more than two classes, so we adopt the one-versus-rest strategy that maps a multi-class label to a binary label vector (one-hot encoding). For the image datasets—namely MNIST [12], CIFAR-10 [10], and SVHN [13]—color images are first transformed to grayscale images, and then the maximum range of each feature is scaled to the interval $[0, 1]$. For the speech dataset TIMIT [7], we normalize each feature by its z-score. For the text dataset 20-Newsgroups [11], we transform each sparse feature vector (bag of words) into a dense feature vector by summing up its corresponding word embeddings obtained from [14].

For each dataset, we subsample a training set (of size n) uniformly at random without replacement. For the 20-Newsgroups dataset, which does not have a test set provided, we randomly pick 1/8 of the full dataset for use as a test set.

Model training. Each model is trained to minimize the squared loss on the given training set. Without regularization, such model is able to interpolate the training set when its capacity surpasses certain threshold (interpolation threshold). For comparison, we report the test/train risk for zero-one and squared loss. In experiments for neural networks and ensemble methods, we repeat the same experiment five times and report the mean for the risks. RFF and Random ReLU experimental results were reported based on a single run as the results were empirically highly consistent.

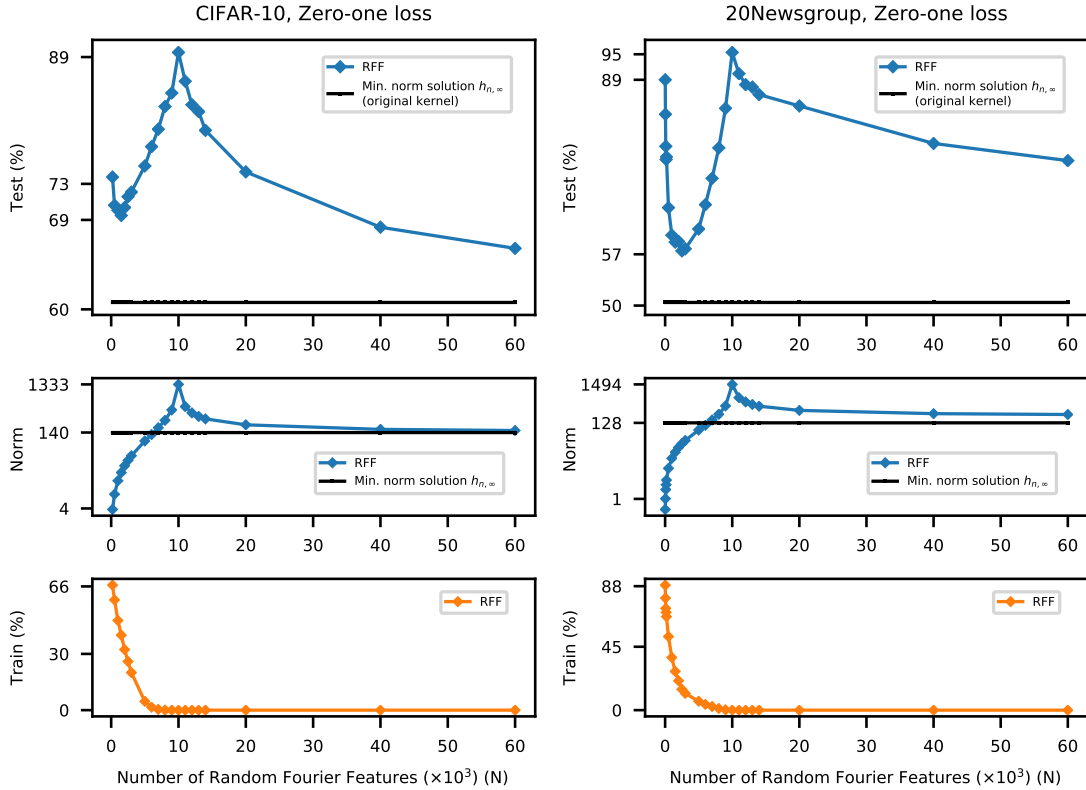


Fig. S1: **Double descent risk curve for RFF model.** Test risks (log scale), coefficient ℓ_2 norms (log scale), and training risks of the RFF model predictors $h_{n,N}$ learned on subsets of CIFAR-10 and 20Newsgroups ($n = 10^4$). The interpolation threshold is achieved at $N = 10^4$.

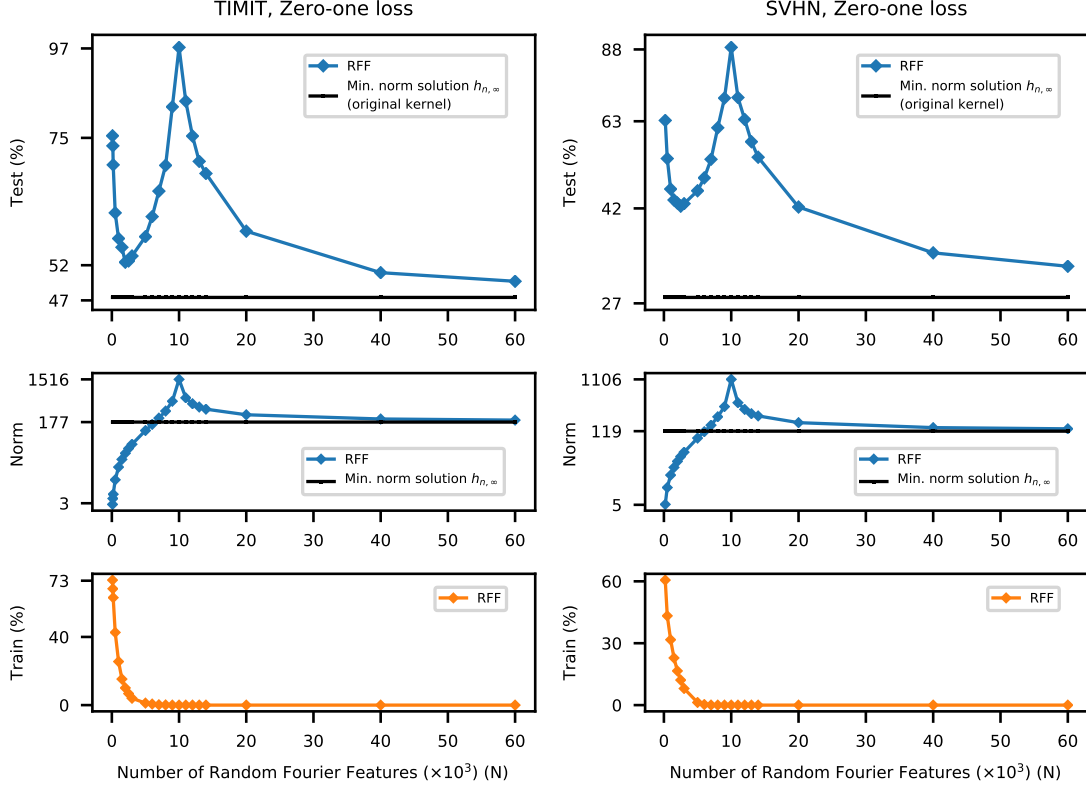


Fig. S2: **Double descent risk curve for RFF model.** Test risks (log scale), coefficient ℓ_2 norms (log scale), and training risks of RFF model predictors $h_{n,N}$ learned on subsets of TIMIT and SVHN ($n = 10^4$). The interpolation threshold is achieved at $N = 10^4$.

S3 Additional experimental results for neural networks

S3.1 Random Fourier Feature models

We provide additional experimental results for several real-world datasets. Fig. S1 illustrates double descent behavior for CIFAR-10 and 20Newsgroup. Fig. S2 shows similar curves of zero-one loss for TIMIT and SVHN. The random feature vectors $v_1, \dots, v_N$ are sampled independently from $\mathcal{N}(0, \sigma^{-2} \cdot I)$, the mean-zero normal distribution in $\mathbb{R}^d$ with covariance $\sigma^{-2} \cdot I$. The bandwidth parameter σ is set to 5, 5, 5, 0.1, and 16 for MNIST, SVHN, CIFAR-10, 20-Newsgroup, and TIMIT, respectively.

S3.2 Random ReLU Feature models

We show that the double descent risk curve also appears with Random ReLU feature networks [3]. Such networks are similar to the RFF models, except that they use the ReLU transfer function. Specifically, the Random ReLU features model family $\mathcal{H}_N$ with N parameters consists of functions $h: \mathbb{R}^d \rightarrow \mathbb{R}$ of the form

$$h(x) = \sum_{k=1}^N a_k \phi(x; v_k) \quad \text{where} \quad \phi(x; v) := \max(\langle v, x \rangle, 0).$$

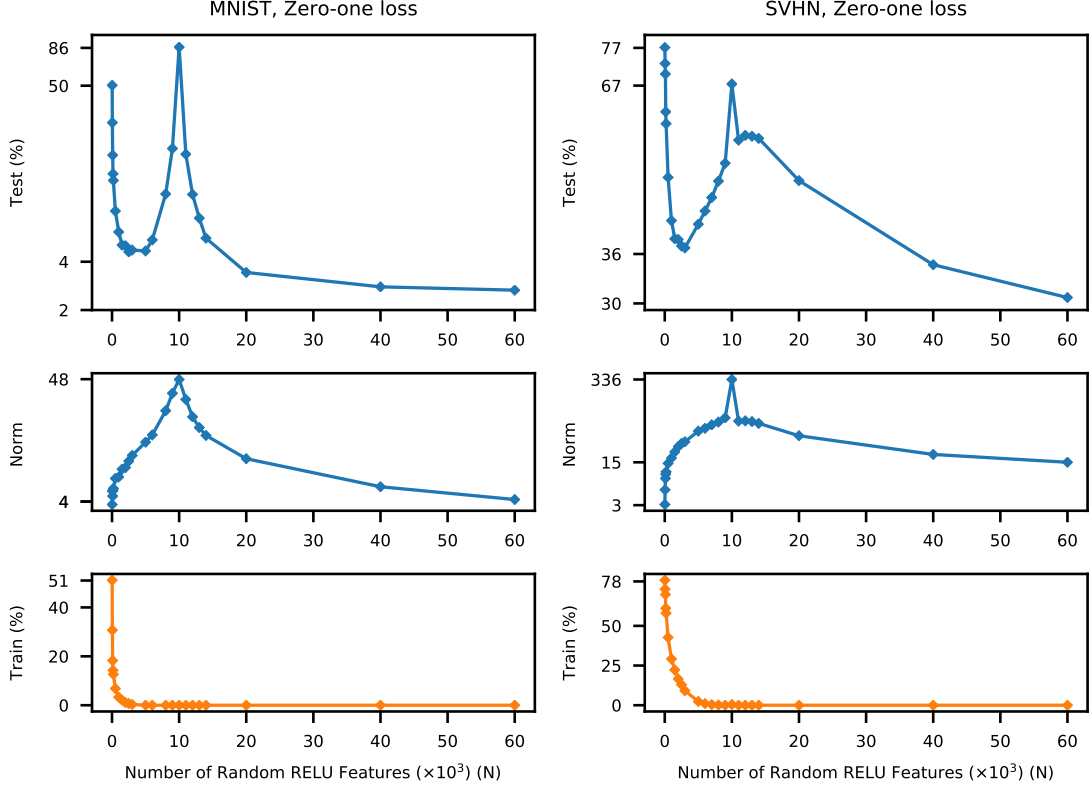


Fig. S3: **Double descent risk curve for Random ReLU model.** Test risks (log scale), coefficient ℓ_2 norms (log scale), and training risks of the Random ReLU Features model predictors $h_{n,N}$ learned on subsets of MNIST and SVHN data ($n = 10^4$). The interpolation threshold is achieved at $N = 10^4$. Regularization of $4 \cdot 10^{-6}$ is added for SVHN to ensure numerical stability near interpolation threshold.

The vectors $v_1, \dots, v_N$ are sampled independently from uniform distribution over surface of unit sphere in $\mathbb{R}^d$. The coefficients a_k are learned using linear regression. Fig. S3 illustrates zero-one loss with Random ReLU features for MNIST and SVHN data. Ridge regularization with parameter $\lambda := 4 \cdot 10^{-6}$ is added in SVHN experiments to ensure numerical stability near the interpolation threshold. For MNIST experiments, no regularization is added. We observe that the resulting risk curves and the norm curves are very similar to those for RFF.

S3.3 Fully connected neural networks

In our experiments, we use fully connected neural networks with a single hidden layer. To control the capacity of function class, we vary the number of hidden units. We use stochastic gradient descent (SGD) to solve the ERM optimization problem in this setting.

The ERM optimization problem in this setting is generally more difficult than that for RFF and ReLU feature models due to a lack of analytical solutions and non-convexity of the problem. Consequently, SGD is known to be sensitive to initialization. To mitigate this sensitivity, we use a “weight reuse” scheme with SGD in the under-parametrized regime ($N < n$), where the parameters obtained from training a smaller neural

network are used as initialization for training larger networks. This procedure, detailed below, ensures decreasing training risk as the number of parameters increases. In the over-parametrized regime ($N \geq n$), we use standard (random) initialization, as typically there is no difficulty in obtaining near-zero training risk.

Additional experimental results for neural networks are shown in Fig. S4. Results for MNIST and CIFAR-10 with weight reuse are reported in Fig. S4A. Results for MNIST without weight reuse are reported in Fig. S4B. In this setting, all models are randomly initialized. While the variance is significantly larger, and the training loss is not monotonically decreasing, the double descent behavior is still clearly discernible.

We now provide specific details below. We use SGD with standard momentum (parameter value 0.95) implemented in [4] for training. In the weight reuse scheme, we assume that we have already trained a smaller network with H_1 hidden units. To train a larger network with $H_2 > H_1$ hidden units, we initialize the first H_1 hidden units of the larger network to the weights learned in the smaller network. The remaining weights are initialized with normally distributed random numbers (mean 0 and variance 0.01). The smallest network is initialized using standard Glorot-uniform distribution [8]. For networks smaller than the interpolation threshold, we decay the step size by 10% after each of 500 epochs, where an epoch denotes a pass through the training data. For these networks, training is stopped after classification error reached zero or 6000 epochs, whichever happens earlier. For networks larger than interpolation threshold, fixed step size is used, and training is stopped after 6000 epochs.

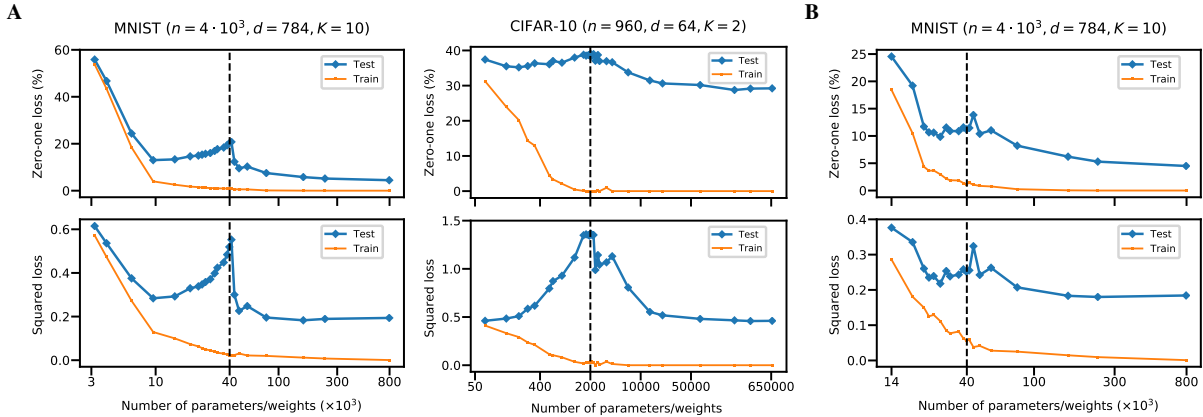


Fig. S4: **Double descent risk curve for fully connected neural networks.** In each plot, we use a dataset with n subsamples of d dimension and K classes for training. We use networks with a single hidden layer. For network with H hidden units, its number of parameters is $(d+1) \cdot H + (H+1) \cdot K$. The interpolation threshold is observed at $n \cdot K$ and is marked by black dotted line in figures. **(A)** Weight reuse before interpolation threshold and random initialization after it. Here we use a subset of CIFAR-10 with 2 classes (cat, dog) and downsampled image features (8×8). **(B)** No weight reuse (random initialization for all ranges of parameters).

S3.4 Synthetic model

We now discuss the nature of the double descent risk curve in the context of a simple synthetic model, which can be viewed as a version of RFF for functions on the one-dimensional circle. Consider the class $\mathcal{H}$ of periodic complex-valued functions on the interval $[0, 2\pi]$, and let

$$e_k(x) := \exp(\sqrt{-1}(k-1)x)$$

for positive integers k . Fix a probability distribution $p = (p_1, p_2, \dots)$ on the positive integers. For each integer N , we generate a random function class $\mathcal{H}_N$ by (i) sampling independently from p until N distinct indices $k_1, \dots, k_N$ are chosen, and then (ii) let $\mathcal{H}_N$ be the linear span of $e_{k_1}, \dots, e_{k_N}$. Here, N is the number of parameters to specify a function in $\mathcal{H}_N$ and also reflects the capacity of $\mathcal{H}_N$.

We generate data from the following model:

$$y_i = h^*(x_i) + \varepsilon_i$$

where the target function $h^* = \sum_k \alpha_k^* e_k$ is in the span of the e_k , and $\varepsilon_1, \dots, \varepsilon_n$ are independent zero-mean normal random variables with variance σ^2 . The $x_1, \dots, x_n$ themselves are drawn uniformly at random from $\{2\pi j/M : j = 0, \dots, M-1\}$ for $M := 4096$. We also let $\alpha_k^* := p_k$ for all k . The signal-to-noise ratio (SNR) is $\mathbb{E}[h^*(x_i)^2]/\sigma^2$.

Given data $(x_1, y_1), \dots, (x_n, y_n) \in [0, 2\pi] \times \mathbb{R}$, we learn a function from the function class $\mathcal{H}_N$ using empirical risk minimization, which is equivalent to ordinary least squares over an N -dimensional space. Interpolation is achieved when $N \geq n$, so in this regime, we choose the interpolating function $h = \sum_{j=1}^N \alpha_{k_j} e_{k_j}$ of smallest (squared) norm $\|h\|_{\mathcal{H}}^2 = \sum_k \alpha_k^2/p_k$.

Our simulations were carried out for a variety of sample sizes ($n \in \{64, 128, 256, 512, 1024, 2048\}$) and are all repeated independently 20 times; our plots show averages over the 20 trials. The results confirm our hypothesized double descent risk curve, as shown in Fig. S5 for $n = 256$; the results are similar for other n . The peak occurs at $N = n$, and the right endpoint of the curve is lower than the bottom of the U curve. The norm of the learned function also peaks at $N = n$ and decreases for $N > n$.

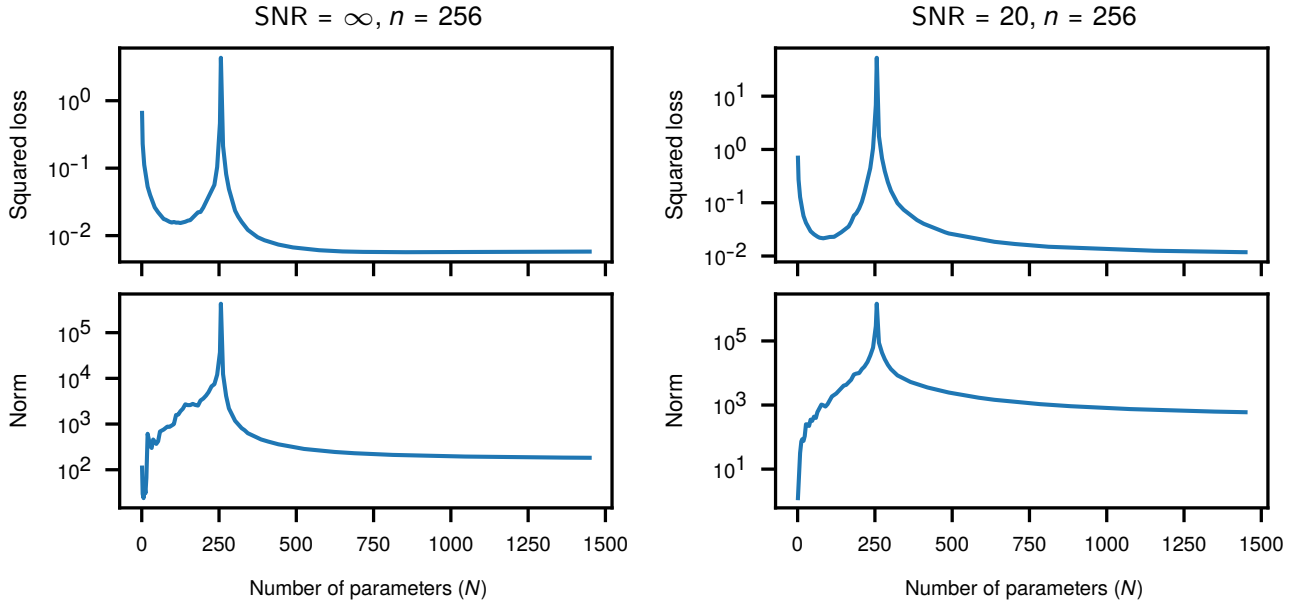
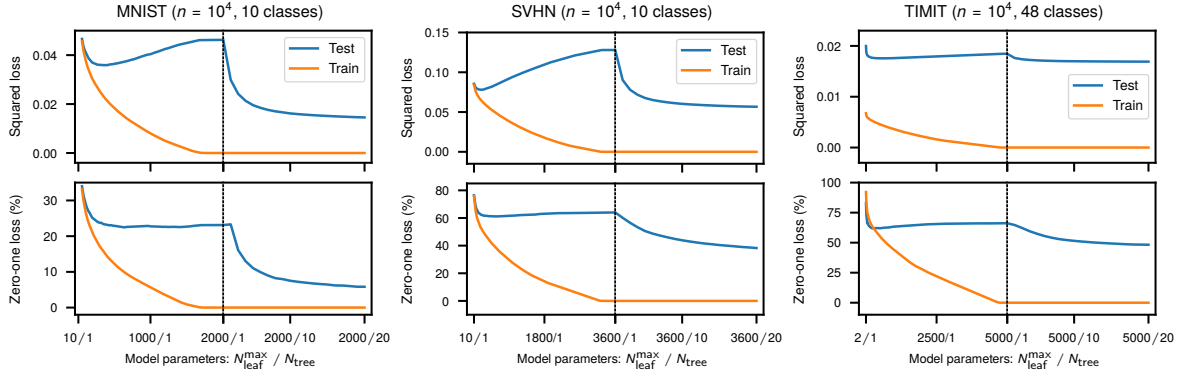


Fig. S5: **Results from the synthetic model at SNR = ∞ and SNR = 20.** Top: excess test risk under squared loss of learned function. Bottom: norm of learned function $\|h\|_{\mathcal{H}_\infty}$. For $n = 256$ training samples, the interpolation threshold is reached at $N = 256$.

A



B

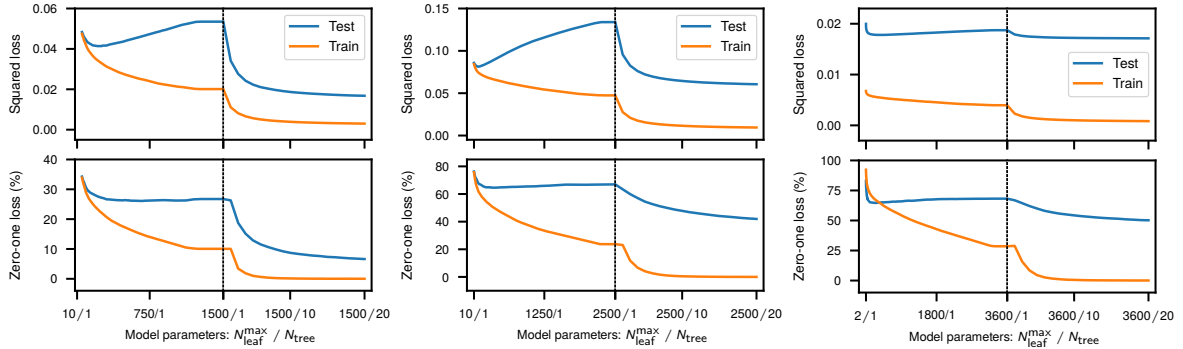


Fig. S6: **Double descent risk curve for random forests** . In all plots, the double descent risk curve is observed for random forest with increasing model complexity on regression tasks. Its complexity is controlled by the number of trees N_{tree} and the maximum number of leaves allowed for each tree $N_{\text{leaf}}^{\text{max}}$. (A) Without bootstrap re-sampling, a single tree can interpolate the training data. (B) With bootstrap re-sampling, multiple trees are needed to interpolate the data.

S4 Additional results with Random Forests

We train standard random forests introduced by Breiman [2] for regression problems. When splitting a node, we randomly select a subset of features whose number is the square root of the number of the total features, a setting which is widely used in mainstream implementations of random forest. We control the capacity of the model class by choosing the number of trees (N_{tree}) and limiting the maximum number of leaves in each tree ($N_{\text{leaf}}^{\text{max}}$). We put minimum constraints on the growth of each tree: there is no limit for the tree depth and we split each tree node whenever it is possible.

To interpolate the training data, we disable the bootstrap re-sampling for results in Fig. S6A, which has been investigated under the name “Perfect random tree ensembles” by Cutler et al. [5]. We see clear double decent risk curve (with both squared loss and zero-one loss) as we increase the capacity of the model class (although the U-shaped curve is less apparent with zero-one loss). In Fig. S6B, we run the same experiments with bootstrap re-sampling enabled, which show similar double decent risk curves.

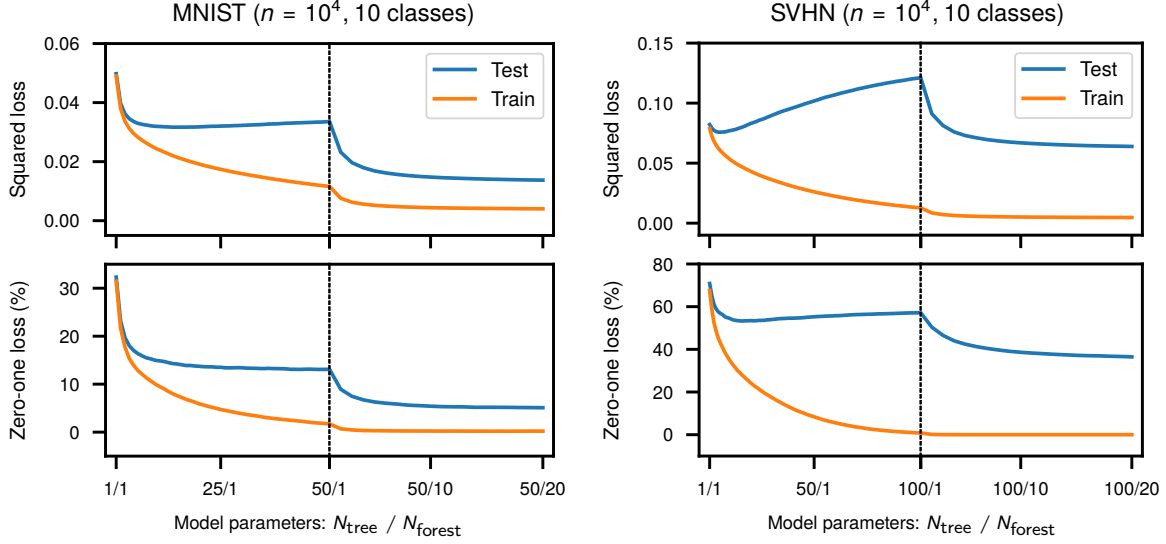


Fig. S7: **Double descent risk curve for L_2 -boosting trees.** In both plots, we increase the model complexity by first increasing the number of boosting (random) trees (N_{tree}) which form a forest, then averaging several such forests (N_{forest}). Each tree is constrained to have no more than 10 leaves. For fast interpolation, the gradient boosting is applied with low shrinkage parameter (0.85).

S5 Results with L_2 -boosting

We now show double descent risk curve for L_2 -boosting (random) trees introduced by Friedman [6]. When splitting a node in a tree, we randomly select a subset of features whose number is the square root of the number of the total features. We constrain each tree to have a small number of leaves (no more than 10). As the number of trees increases, the boosted trees gradually interpolate the training data and form a forest. To quickly reach interpolation, we adopt low shrinkage (parameter value 0.85) for gradient boosting. To go beyond the interpolation threshold, we average the predictions of several such forests which are randomly constructed and trained with exactly same hyper-parameters. The capacity of our model is hence controlled by the number of forests (N_{forest}) and the number of trees (N_{tree}) in each forest.

Fig. S7 shows the change of train and test risk as the model capacity increases. We see the double descent risk curve for both squared loss and zero-one loss. We also observe strong over-fitting under squared loss before the interpolation threshold. For similar experiments with high shrinkage (parameter value 0.1), the double descent risk curve becomes less apparent due to the regularization effect of high shrinkage [9].

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